✈ AIR UNIT #8 — LiDAR

ID:

obs-air-lidar

Role: 3D Terrain Mapping / Structural Scanning

Tier: Advanced

Status: Active

● PHYSICAL & TECHNICAL

Length: 2.8 m
Width: 2.1 m
Height: 1.0 m
Mass: 28 kg
Power Source: High-capacity battery
Energy Capacity: 2600 Wh
Energy Consumption: 60 Wh/min
Endurance: 30–45 min
Payload Capacity: 2.0 kg
Modules: LiDAR emitter array
Navigation System: GPS + LiDAR-assisted stabilization
Structural Integrity: 140 HP
Armor Rating: 0.20
Heat Signature: Medium
Noise Signature: Medium
Top Speed: 20 m/s
Acceleration: Medium
Turning Radius: Tight
Climb Rate: Medium
Stability Rating: Very High (mapping precision)

● AUTONOMY & AI BEHAVIOR

Behavior Profile: Mapping / Survey

Aggression: 0.0
Caution: 0.8
Persistence: 0.9
Flanking Likelihood: 0.0
Retreat Threshold: 0.7
Target Priority: Terrain features / structures
Formation Discipline: High
Risk Tolerance: Low

Sensors

Visual Range: 1400 m
Thermal Range: 600 m
Acoustic Range: Low
Radar Range: None
LiDAR Range: 2000 m (high-resolution)
Signal Detection Range: Medium

Communication

Mesh Range: 6 km
Relay Capacity: 0
Command Link Type: Mapping Grid
Network Priority: High

● COMBAT & THREAT MODEL

Firepower Rating: 0
Accuracy: N/A
Stabilization: Very High
Weapon Mounts: None
Weapon Types: None
Suppression Value: 0
Tracking Strength: Medium
Disruption Strength: Low
Breach Capability: None
Countermeasure Rating: Low

Threat Modeling

Threat Level: 3
Engagement Distance: N/A
Preferred Terrain: Mapping corridors / open air
Preferred Targets: None
Vulnerability Profile: Ballistic / weather

● SQUAD & COMMAND INTEGRATION

Squad Role: 3D Mapping
Squad Size Contribution: 1
Formation Behavior: Grid-pattern scanning
Intent Interpretation Rules: Maintain mapping altitude and coverage
Retreat Behavior: Medium
Breach Behavior: None
Flanking Behavior: None
Reinforcement Behavior: Supports recon and engineering squads

Command Hierarchy

Parent Command Unit: Cartographer / Insight
Subordinate Units: None
Relay Nodes: None
Network Dependencies: Mapping Grid

● VR OPERATOR MODE

Operator Seat Type: Drone View
Operator HUD: LiDAR Grid Overlay
Camera Feeds: Optical
Thermal View: Yes
LiDAR View: Yes (full-resolution)
Sonar View: No
Tactical Overlay: Terrain mesh visualization

Operator Controls

Movement Controls: Full drone flight
Sensor Controls: LiDAR sweep / density adjustment
Ability Controls: Scan, Map Terrain, Structural Sweep
Weapon Controls: None

Stabilization Controls: Auto-hover, mapping-lock

Operator Abilities

LiDAR Sweep

Terrain Mesh Generation

Structural Scan

Mark Terrain Features

⚫ ECONOMY & LOGISTICS

Build Cost: 220 Metal / 200 Electronics

Build Time: 70 sec

Factory Type: Drone Bay

Tech Requirements: Mapping Systems I

Maintenance Cost: Medium

Repair Time: 18 sec

Supply Consumption: Medium

Operational Cost: Moderate

Deployment Requirements: Standard drone pad

Transport Compatibility: All aerial racks

Launch Method: Vertical takeoff

🔧 BUILD REQUIREMENTS (PLAYER COLLECTION SYSTEM)

Common Components

16× **Medium Alloy Plates**

12× **Composite Frame Rods**

8× **Micro-Servo Actuators**

4× **Stabilizer Fins**

1× **LiDAR Emitter Housing**

Electronics

16× **Circuit Bundles**

12× **Signal Processors**

6× **Drone Control Chips**

4× **LiDAR Processing Chips**

Power

1× **High-Capacity Battery Pack**
3× **Power Regulation Nodes**

Rare / Tradeable

1× **High-Resolution LiDAR Core**
1× **Mapping Grid Microprocessor**

Optional Upgrades

Extended LiDAR Range Module
High-Density Scan Grid
Anti-Jamming Shielding
Silent Rotor Kit